

SUMEDH GAIKWAD

Backend Software Engineer | Full Stack Development | AI/ML Integration
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SUMMARY

Backend focused Software Engineer with full stack experience building and deploying database backed applications and AI driven systems. Published researcher in reinforcement learning applied to intelligent transportation systems, with hands on experience across backend architecture, APIs, PostgreSQL, and AI/LLM integration.

PROJECTS

Finance One Personal Financial Copilot

[Live Demo](#)

Tech: React, TypeScript, Vite, TanStack Query, Node.js, Express, Prisma, PostgreSQL, Zod, Ollama/LLMs

- Full-stack finance app with JWT auth, transaction CRUD, budget locking, categories, and dashboard analytics, TanStack Query handles server-state, caching, mutations, and invalidation.
- Working intent classification pipeline converts natural language into validated TransactionQuery inputs for deterministic Prisma/PostgreSQL execution.
- LLM layer strictly separated from financial business logic broader AI copilot in active development. MVP 1 · Live

Atrio Real-Time Collaborative Workspace

[Atrio.sumedhgaikwad.com](#)

Tech: React, Node.js, Express, MongoDB, Socket.io, JWT

- Real-time shared notes/task workspace with Socket.io live sync and workspace-scoped broadcasts REST API handles persistence and business logic.
- Addressed reconnection edge cases and concurrent/out of order events; JWT auth for secure multi user sessions deployed and used by real users during beta.
- Designed a hybrid REST + Socket.io architecture, keeping persistence and business logic in the API while using workspace scoped sockets for Real time event propagation.

Virtual2Reality Real Estate Discovery Platform · Client Project

[virtual2reality.in](#)

Tech: React, TypeScript, Node.js, Express, PostgreSQL, Prisma, Cloudinary

- Designed and implemented a Developer → Project → Configuration domain model supporting property discovery, contextual media, enquiries, and administrative management delivered end-to-end for a real-estate client.
- Designed a layered PostgreSQL backend with a smart query builder for property discovery, backed by validation, security, indexing, and data integrity controls.
- Delivered the platform end to end from requirements analysis through implementation and deployment, working with client requirements while maintaining separate public discovery and administrative workflows.

PROFESSIONAL EXPERIENCE

Software Engineering Intern Actima Software, Pune

Jan 2025 – Aug 2025

Tech: TypeScript, React, Prisma, PostgreSQL, Docker

- Delivered end to end vertical slices of client requested features: requirements analysis, implementation, database integration, testing, and deployment.
- Built a feature for clients to retrieve emails by time ranges and constraints participated in client communication and requirement analysis.
- Gained Docker and deployment exposure alongside experienced engineers contributed to Git workflows, code reviews, and iterative delivery.

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, SQL

Frontend: React, Vite, React Router, TanStack Query

Backend & Databases: Node.js, Express, REST APIs, Prisma, PostgreSQL, MongoDB, FastAPI

AI / ML: Reinforcement Learning, PyTorch, NLP, Transformers, LLM Integration, Local LLM Inference, AI Query Pipelines

Tools & Infrastructure: Git, GitHub, Docker, Postman, Ollama/LLMs

EDUCATION

B.E. Computer Engineering, Honors in Artificial Intelligence & Machine Learning

2022 – 2026

Savitribai Phule Pune University · CGPA: 8.83/10 · Final-year SGPA: 9.47/10

RESEARCH & LEADERSHIP

Adaptive Urban Traffic Signal Optimization Using Reinforcement Learning

Published in IJIRT, 2026

- D-DQN RL agent built with PyTorch, simulated via SUMO/TraCI across two topologies; parallel RL vs fixed-time simulation, emergency vehicle preemption, and FastAPI + React + WebSocket monitoring dashboard.
- 58.3% lower average waiting time and 43.8% higher average speed at a single intersection; 12.3% waiting time reduction on an urban arterial grid.
- Evolved from SIH 2025 hackathon entry → Final-year B.E project → peer-reviewed publication (IJIRT Vol. 12, Issue 11, April 2026).

Smart India Hackathon 2025 — Team Lead, Team VEIG_1

- Led Team VEIG_1 through multiple screening rounds to college level selection built and presented the Smart Traffic System MVP, which evolved into a final year project and a peer reviewed publication.